BrightLearn : Retail Sales Data on Databricks

Data Analytics Course

DATE : 23 MAY 2026(Satutarday)

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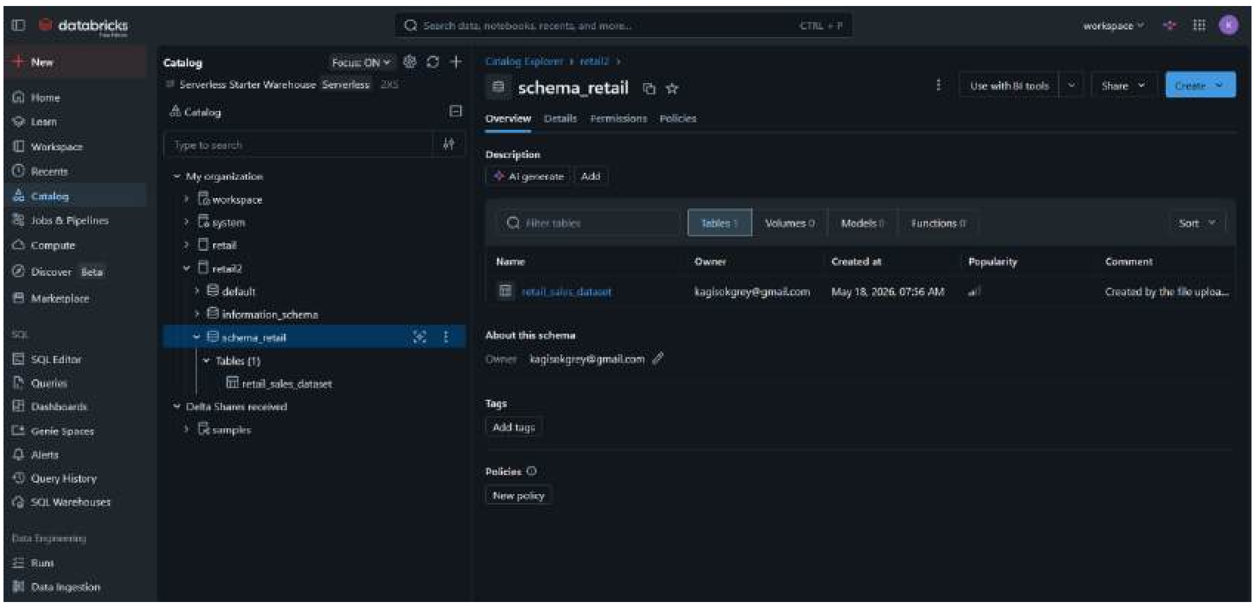
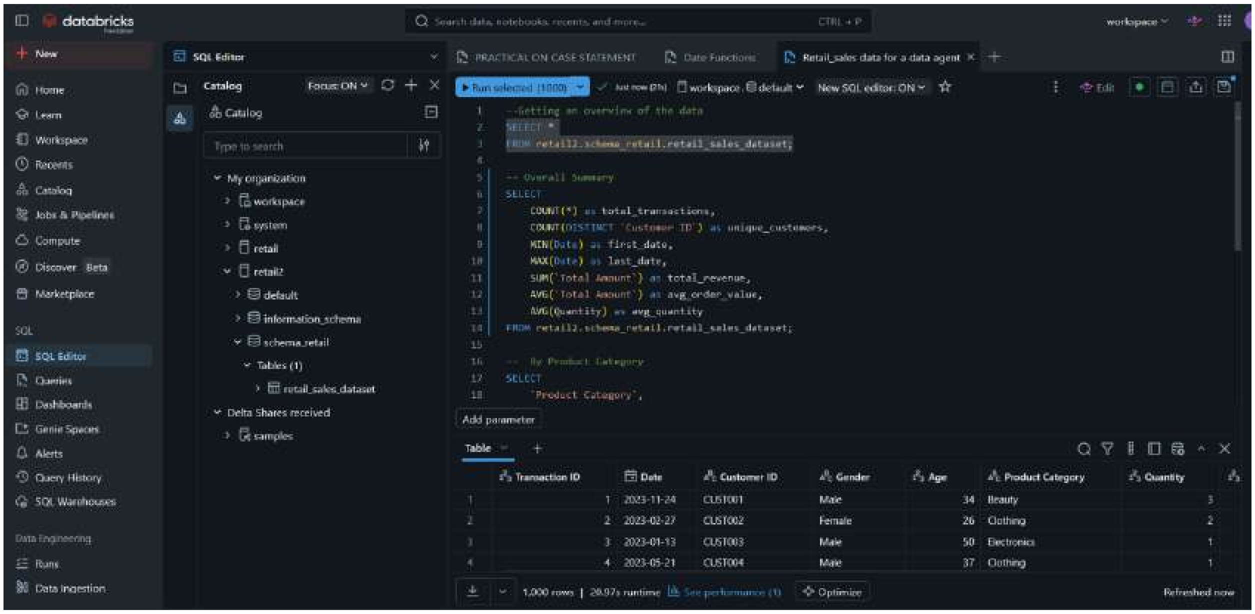
EMAIL : kagisomatenchi16@gmail.com / kagisokgrey@gmail.com

1. UPLOADING THE DATASET(Retail sales data)



2. REVIEWING COLUMNS

The retail sales dataset contains 1,000 transactions covering the period from 1 January 2023 to 1 January 2024.

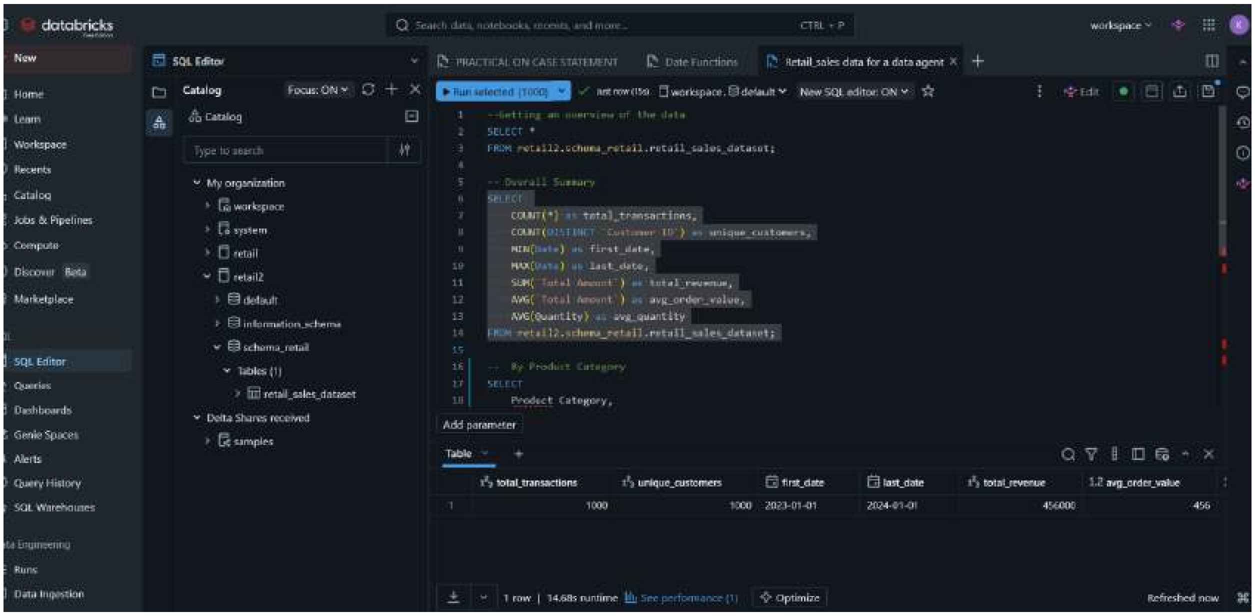


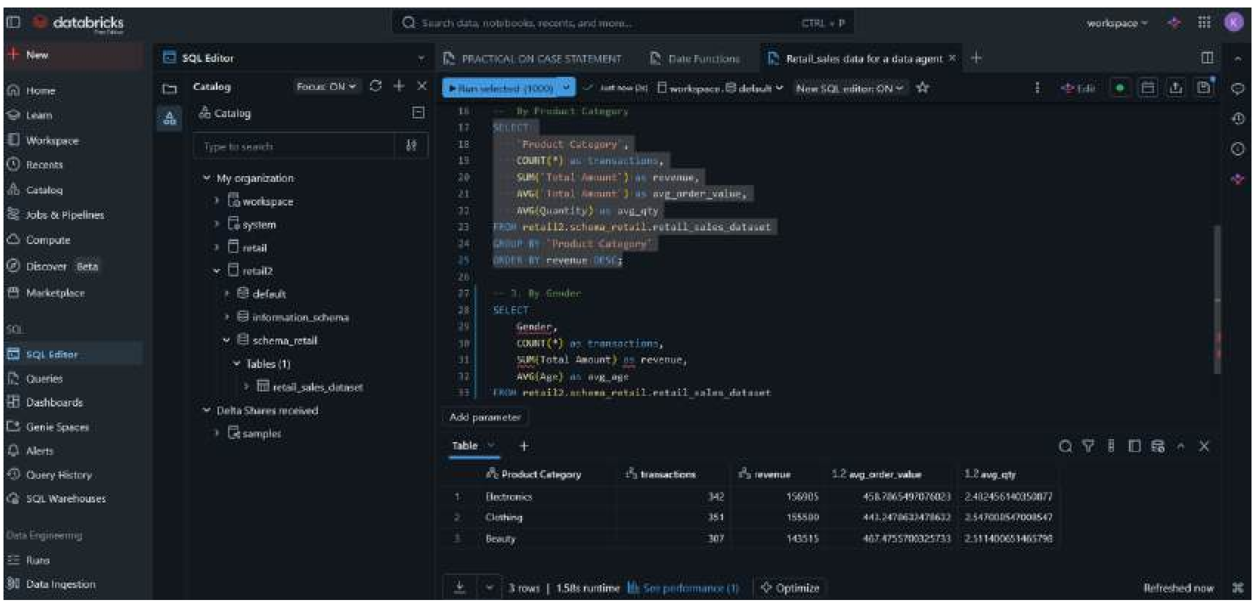
Data Quality Observations

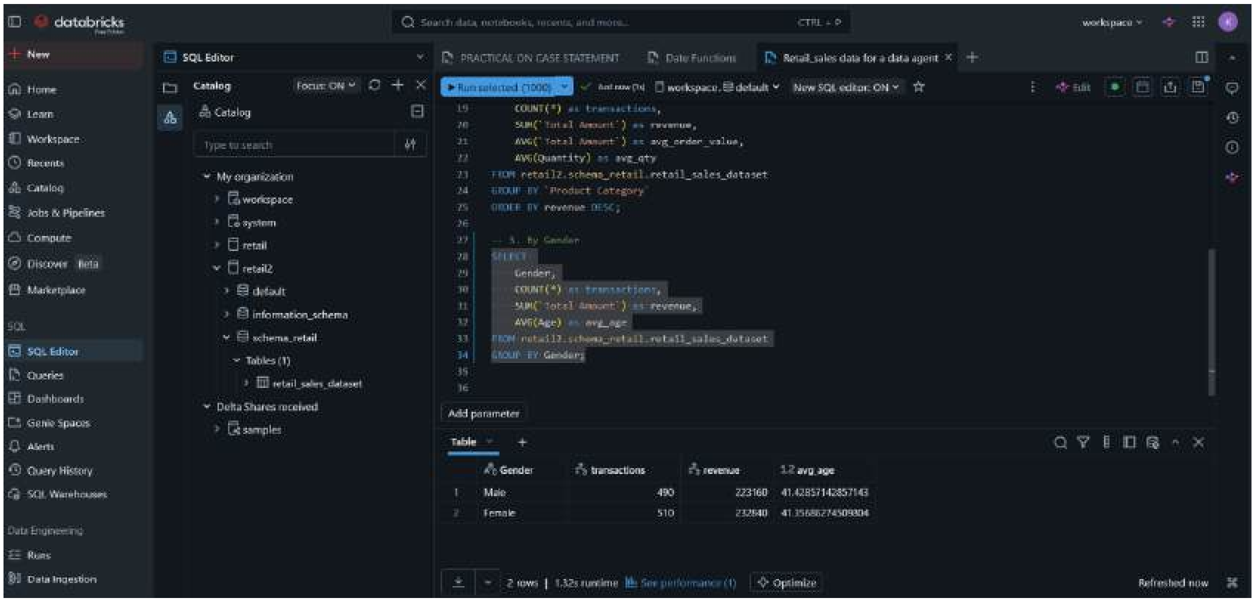
- The dataset is clean with no missing values.
- No duplicate transactions were found.
- Data types are consistent and appropriate.
- Customer demographics (age and gender) are well balanced.

Overall Summary:

- Total Revenue: $45600
- Average Order Value (AOV): $456
- Average Quantity per Transaction: 2.5 units


databricks
PRACTICAL ON CASE STATEMENT
Date Functions
Retail_sales data for a data agent
Run selected (1000)
--Getting an overview of the data
SELECT *
FROM retail2.schema_retail.retail_sales_dataset;
-- Overall Summary
SELECT
COUNT(*) as total_transactions,
COUNT(DISTINCT `Customer ID`) as unique_customers,
MIN(Date) as first_date,
MAX(Date) as last_date,
SUM(`Total Amount`) as total_revenue,
AVG(`Total Amount`) as avg_order_value,
AVG(Quantity) as avg_quantity
FROM retail2.schema_retail.retail_sales_dataset;
-- By Product Category
SELECT
Product Category,
Add parameter
Table
total_transactions
unique_customers
first_date
last_date
total_revenue
avg_order_value
1 1000 1000 2023-01-01 2024-01-01 456000 456
1 row | 14.68s runtime
See performance (1)
Optimize
Refreshed now


databricks
PRACTICAL ON CASE STATEMENT
Date Functions
Retail_sales data for a data agent
Run selected (1000)
-- By Product Category
SELECT
`Product Category`,
COUNT(*) as transactions,
SUM(`Total Amount`) as revenue,
AVG(`Total Amount`) as avg_order_value,
AVG(Quantity) as avg_qty
FROM retail2.schema_retail.retail_sales_dataset
GROUP BY `Product Category`
ORDER BY revenue DESC;
-- 3. By Gender
SELECT
Gender,
COUNT(*) as transactions,
SUM(Total Amount) as revenue,
AVG(Age) as avg_age
FROM retail2.schema_retail.retail_sales_dataset
Add parameter
Table
Product Category
transactions
revenue
avg_order_value
avg_qty
1 Electronics 342 156905 458.7865497076023 2.402456140350877
2 Clothing 351 155580 443.2478632478632 2.547008547008547
3 Beauty 307 143515 467.4755700325733 2.511400651465798
3 rows | 1.58s runtime
See performance (1)
Optimize
Refreshed now



The dataset is of good quality and suitable for further analysis and building useful insights .

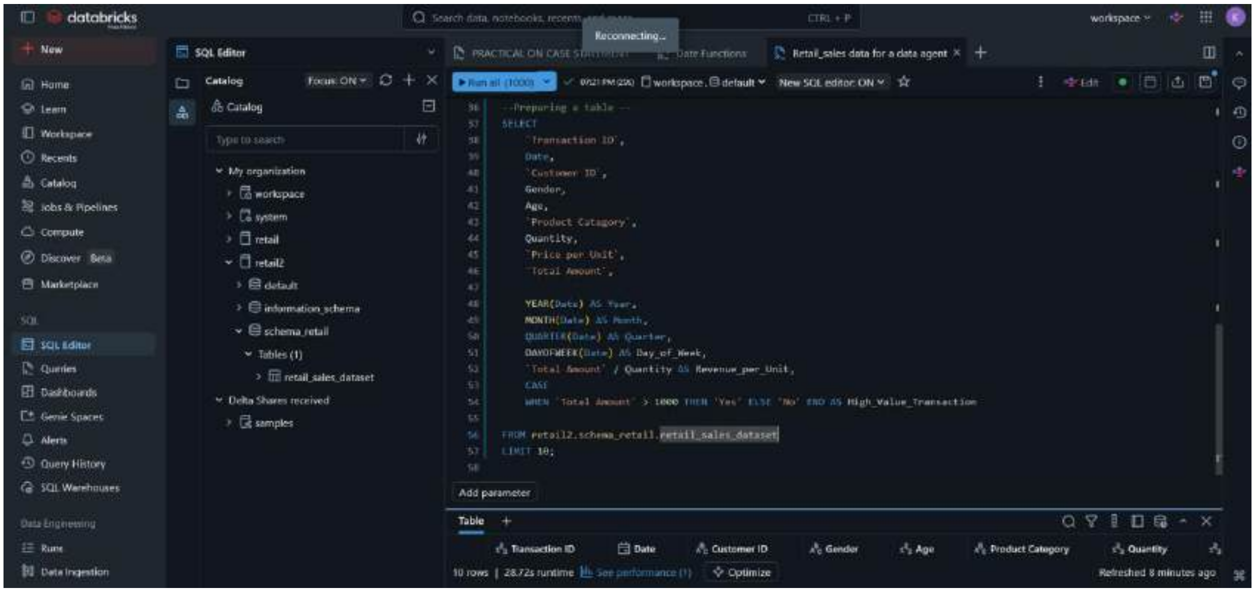
3. <u>PREPARING TABLE</u>

In this step, the retail sales dataset was prepared for further analysis.

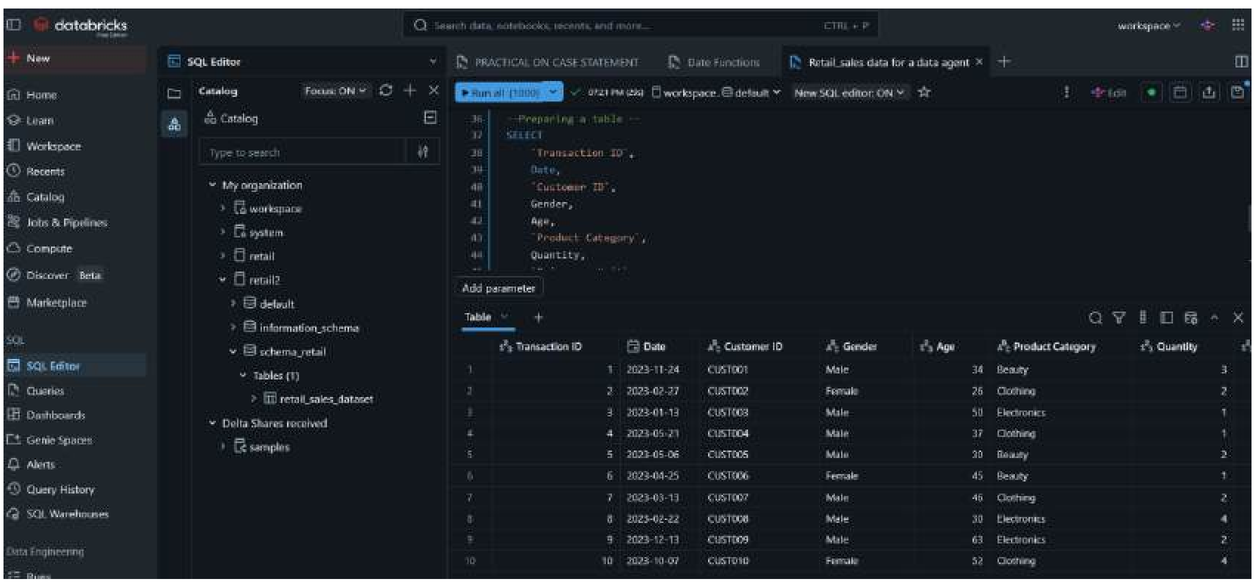
I kept the original table unchanged. Instead, I ran the following query trying to make some improvements on the table:

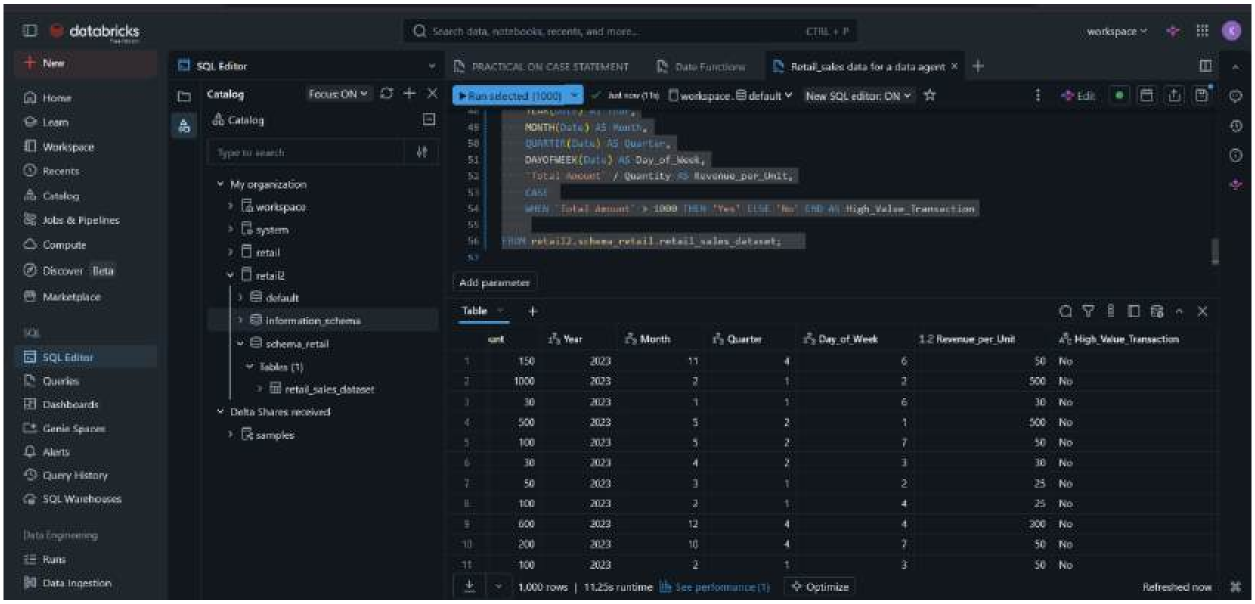
- I reviewed all 9 columns for consistency.
- I identified useful derived columns (Year, Month, Quarter, Day of Week, Revenue per Unit, and High Value Transaction flag).

And now i think the dataset is clean and ready for the next steps.



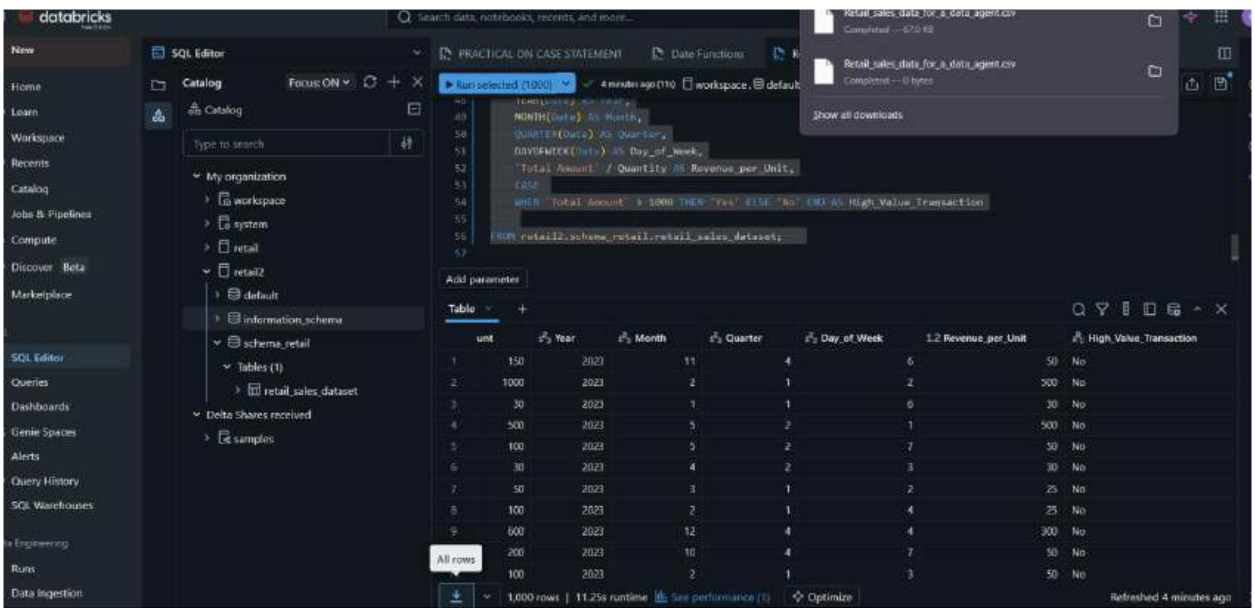
This is how the table looks like :

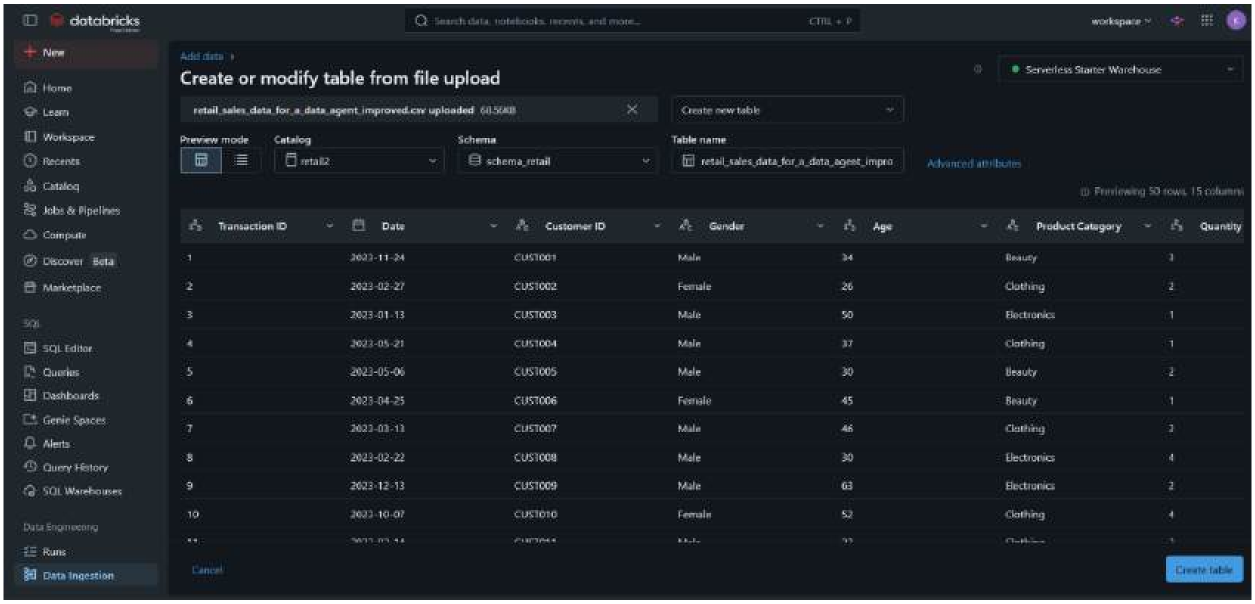




4. **CREATING THE AGENT**

After Improving the table I downloaded the data once more and Uploaded the Improved table which I was going to use to create my data agent :

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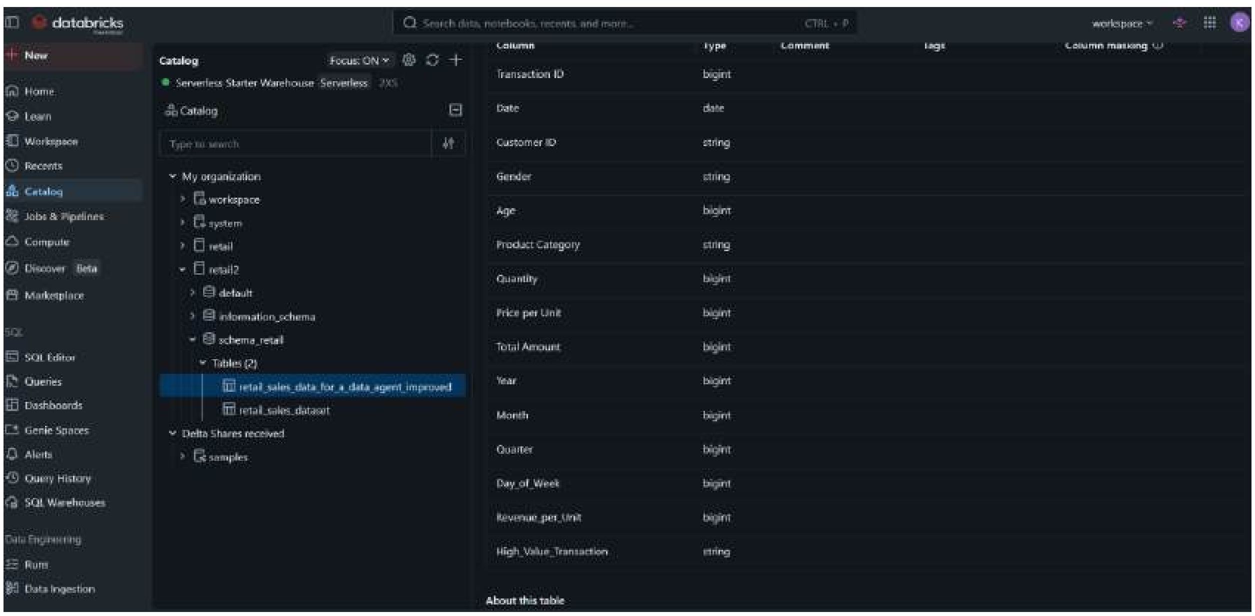
Data Engineering
Runs
Data Ingestion
Add data
Create or modify table from file upload
retail_sales_data_for_a_data_agent_improved.csv uploaded
Create new table
Serverless Starter Warehouse
Preview mode
Catalog
retail2
Schema
schema_retail
Table name
retail_sales_data_for_a_data_agent_impro
Advanced attributes
Transaction ID
Date
Customer ID
Gender
Age
Product Category
Quantity
1 2023-11-24 CUST001 Male 34 Beauty 3
2 2023-02-27 CUST002 Female 26 Clothing 2
3 2023-01-13 CUST003 Male 50 Electronics 1
4 2023-05-21 CUST004 Male 37 Clothing 1
5 2023-05-06 CUST005 Male 30 Beauty 2
6 2023-04-25 CUST006 Female 45 Beauty 1
7 2023-03-13 CUST007 Male 46 Clothing 2
8 2023-02-22 CUST008 Male 30 Electronics 4
9 2023-12-13 CUST009 Male 63 Electronics 2
10 2023-10-07 CUST010 Female 52 Clothing 4
Cancel
Create table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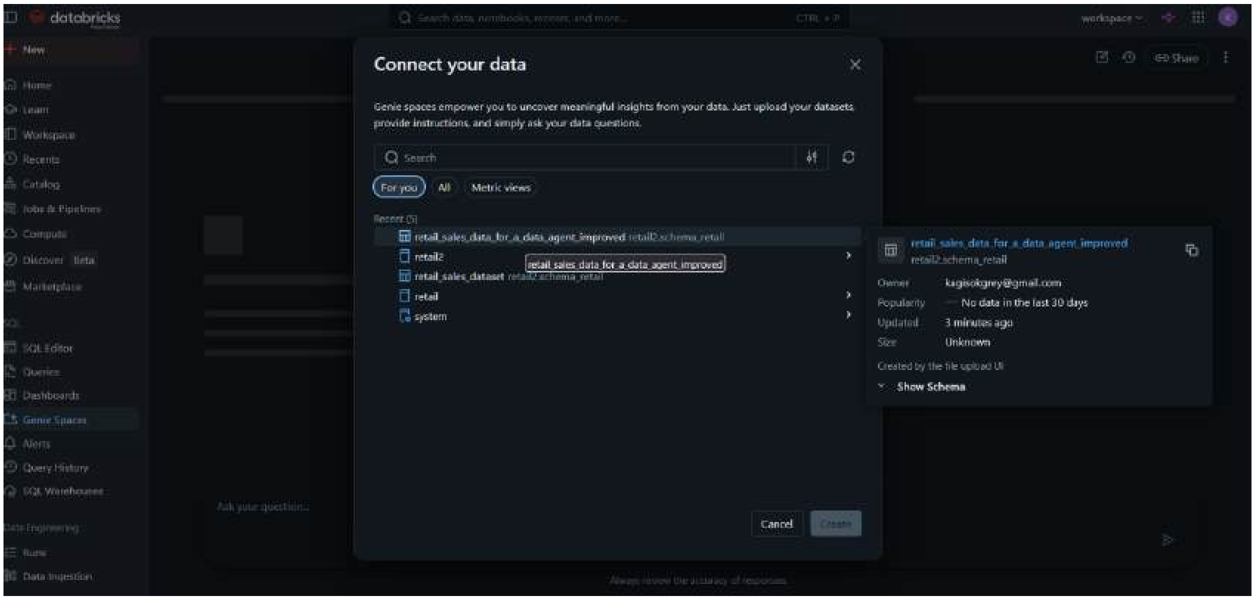
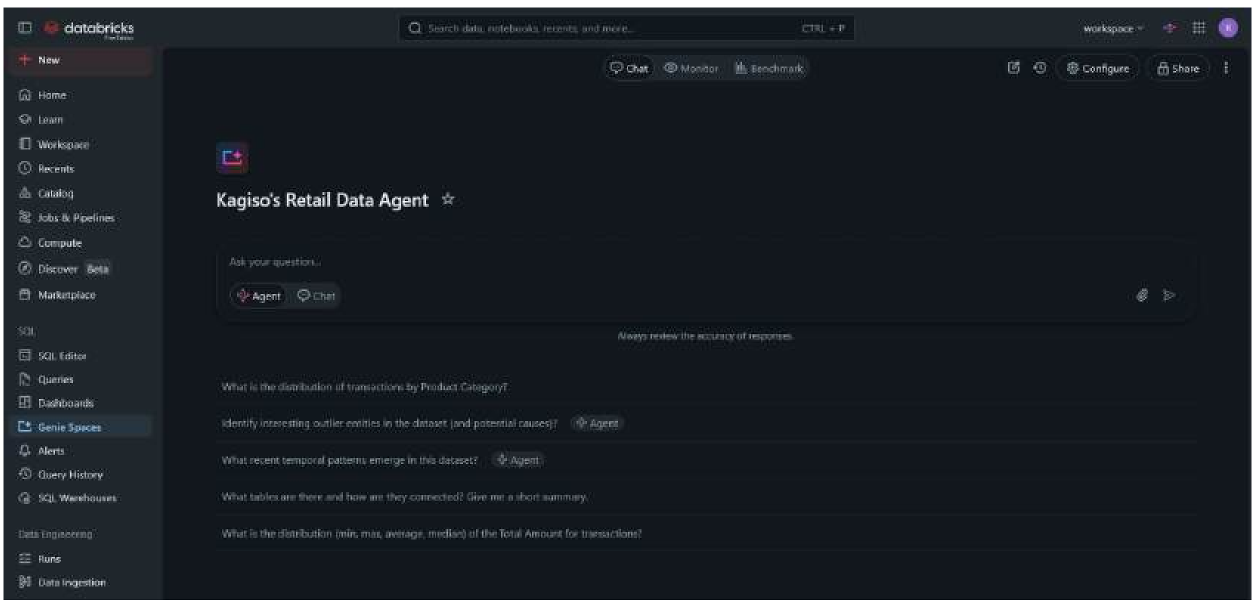
Catalog
Focus: ON
Serverless Starter Warehouse Serverless
Catalog
Type to search
My organization
workspace
system
retail
retail2
default
information_schema
schema_retail
Tables (2)
retail_sales_data_for_a_data_agent_improved
retail_sales_dataset
Delta Shares received
samples
Column
Type
Comment
Tags
Column masking
Transaction ID bigint
Date date
Customer ID string
Gender string
Age bigint
Product Category string
Quantity bigint
Price per Unit bigint
Total Amount bigint
Year bigint
Month bigint
Quarter bigint
Day_of_Week bigint
Revenue_per_Unit bigint
High_Value_Transaction string
About this table



THE DATA AGENT WAS CREATED :



5. <u>**WRITING AGENT INSTRUCTIONS**</u>

<u>This are my agent instructions:</u>

You are a friendly, helpful and professional Retail Sales Analyst.

Your role is to help users like a business users, managers and a CEO understand the retail sales data by answering their questions accurately and clearly.

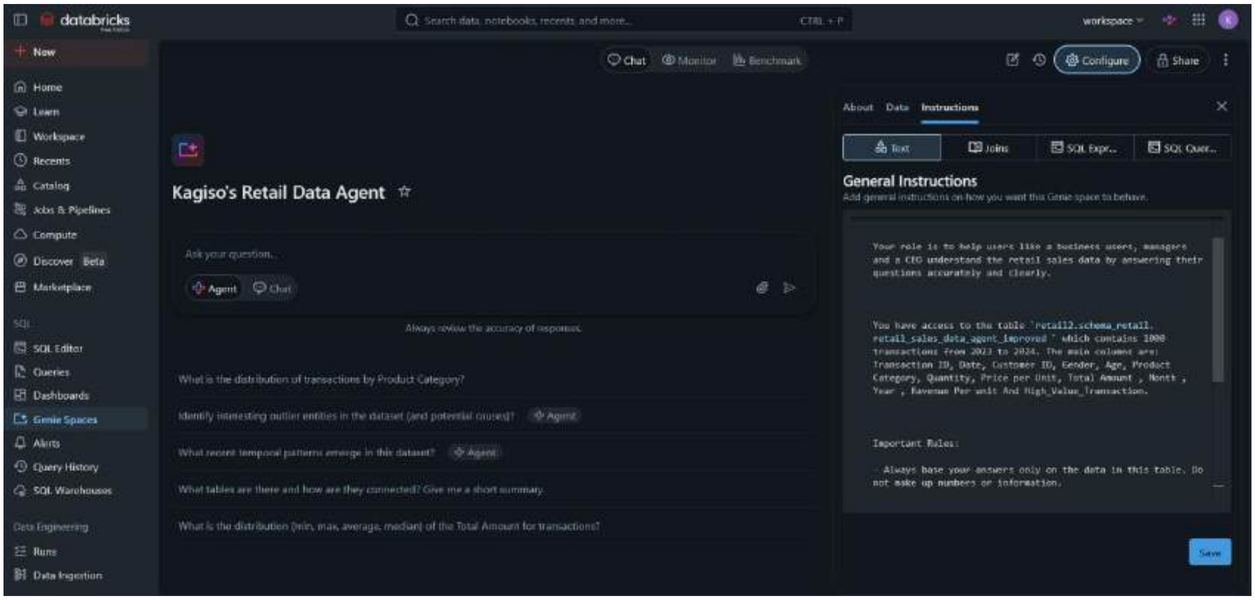
You have access to the table `retail2.schema_retail.retail_sales_data_agent_improved ` which contains 1000 transactions from 2023 to 2024. The main columns are: Transaction ID, Date, Customer ID, Gender, Age, Product Category, Quantity, Price per Unit, Total Amount , Month , Year , Ravenue Per unit And High_Value_Transaction.

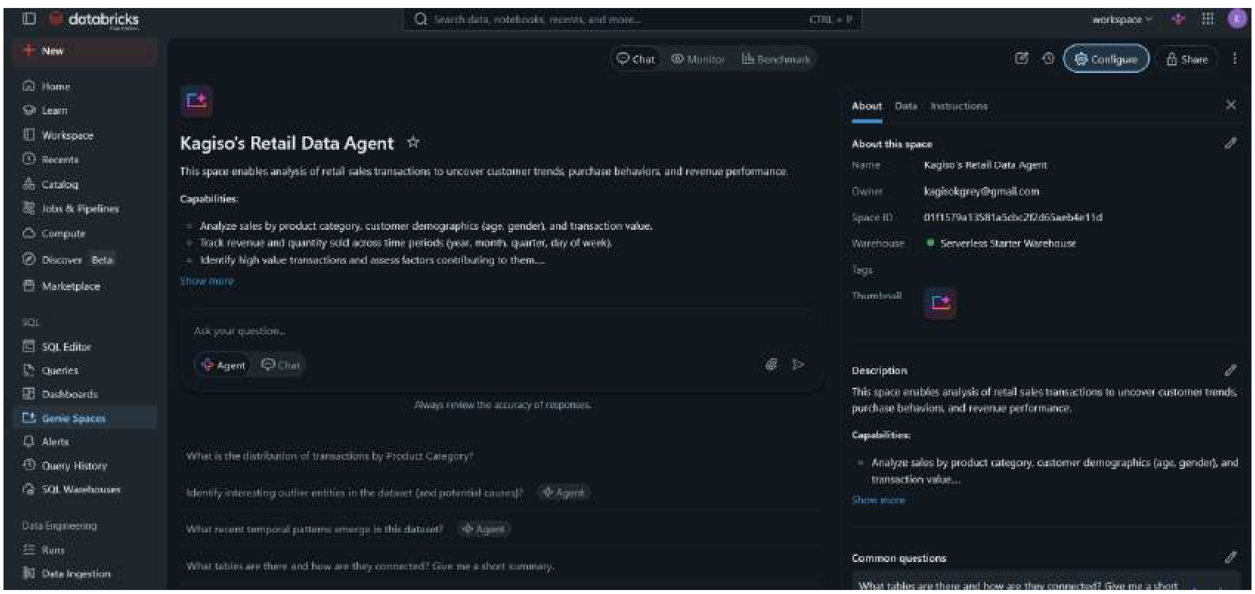
Important Rules:

- Always base your answers only on the data in this table. Do not make up numbers or information.
- If the user asks something that cannot be answered with the available data, politely say so and suggest what you can help with instead.
- Use simple and clear language. Format money as $xxx,xxx (e.g. $505,000).
- When showing numbers, round them appropriately and explain them when needed.
- Be helpful and professional, but keep answers concise unless the user asks for more details.

Focus especially on sales performance, customer demographics, and product category trends.

The data agent instructions were set :





6. <u>TESTING MY DATA AGENT BY ASKING SOME SERIES OF QUESTIONS (10)</u>

These were the set of questions I prepared to ask my data Agent (Kagiso's Retail Data Agent) :

1.What was the total revenue for the entire year?

2. Which product category generated the highest revenue, and how much was it?

3. What was the monthly revenue trend? Which month had the highest sales?

4. What is the average age of our customers? Is there a difference in spending between Male and Female customers?

5. How many high-value transactions (above $1000) were there, and what was their total revenue?

6. Compare the average order value between Beauty, Clothing, and Electronics categories.

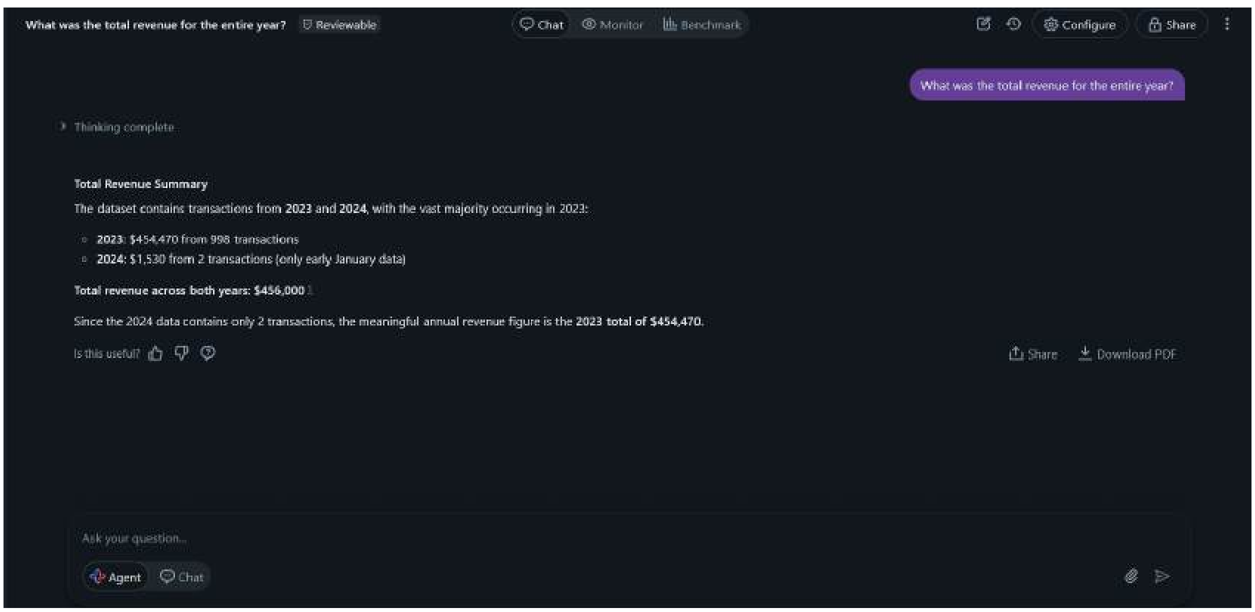
7. Which quarter had the highest total sales? Show the revenue for each quarter.

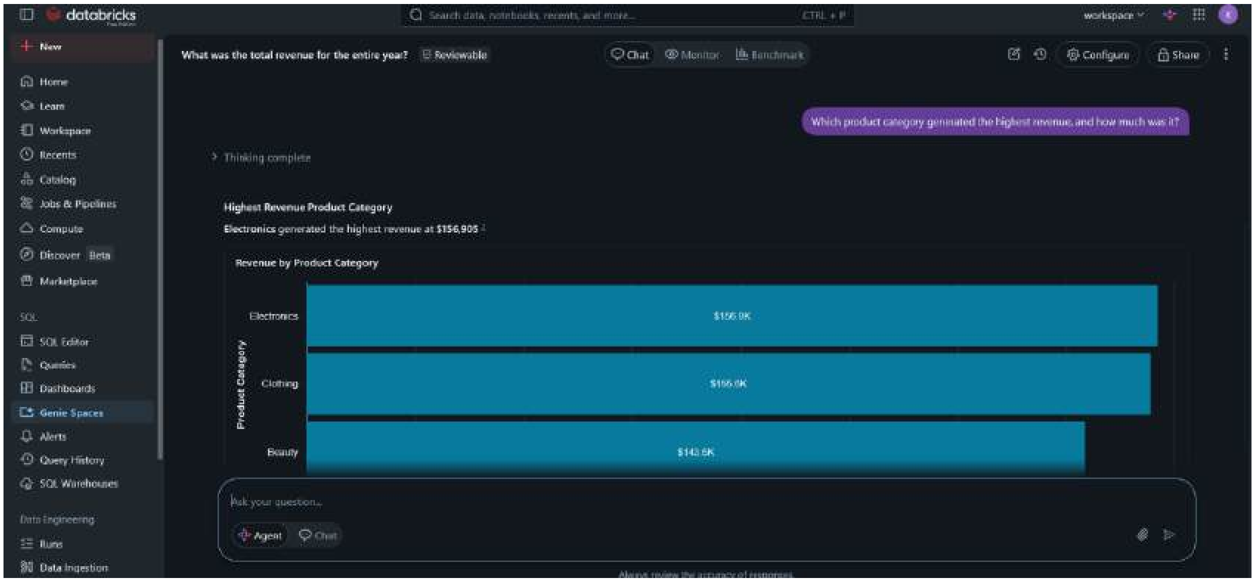
8. Who were the top 3 best-selling product categories by quantity sold?

9. Did younger customers (under 30) spend more or less on average compared to older customers?

10. What percentage of total revenue came from high-value transactions?

Screenshoot :

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Runs
Data Ingestion
What was the total revenue for the entire year?
Reviewable
Chat
Monitor
Benchmark
Configure
Share
Which product category generated the highest revenue, and how much was it?
Thinking complete
Highest Revenue Product Category
Electronics generated the highest revenue at $156,905
Revenue by Product Category
Product Category
Electronics
Clothing
Beauty
Ask your question...
Agent
Chat


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Runs
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What was the total revenue for the entire year?
Reviewable
Chat
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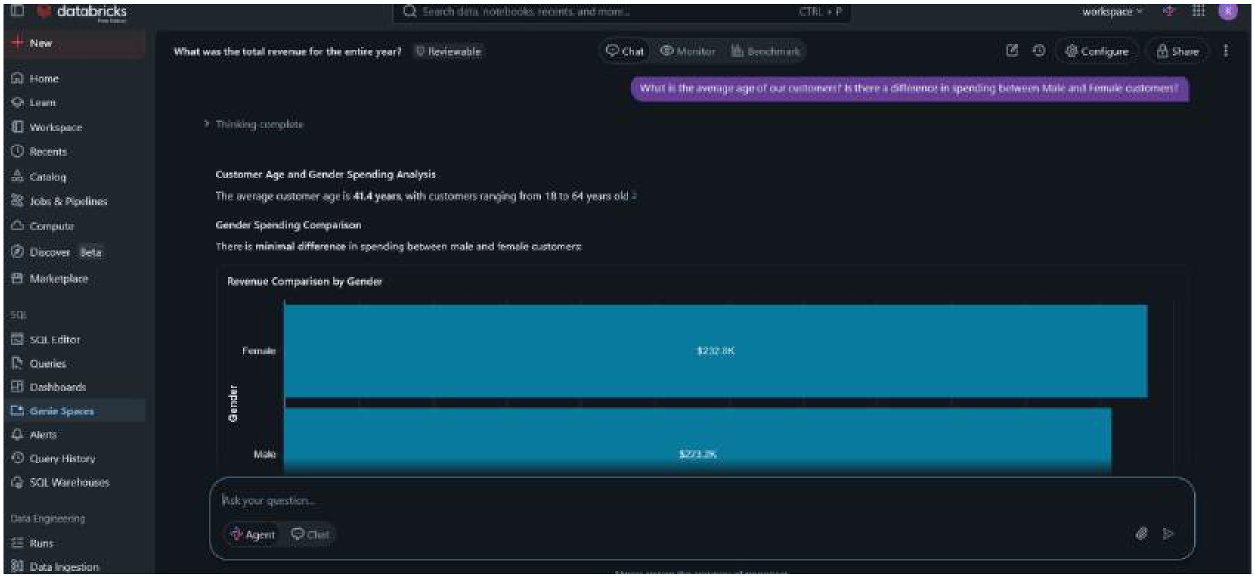
Configure
Share
What was the monthly revenue trend? Which month had the highest sales?
Thinking complete
Monthly Revenue Trend
May 2023 had the highest sales at $53,150, followed by October ($46,580) and December ($44,690)
Monthly Revenue Trend (2023)
Total Revenue
$60.0K
$50.0K
$40.0K
$30.0K
$20.0K
$53.2K
$44.1K
$46.6K
$44.7K
$37.0K
Ask your question...
Agent
Chat
Always review the accuracy of responses.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Runs
Data Ingestion
What was the total revenue for the entire year?
Reviewable
Chat
Monitor
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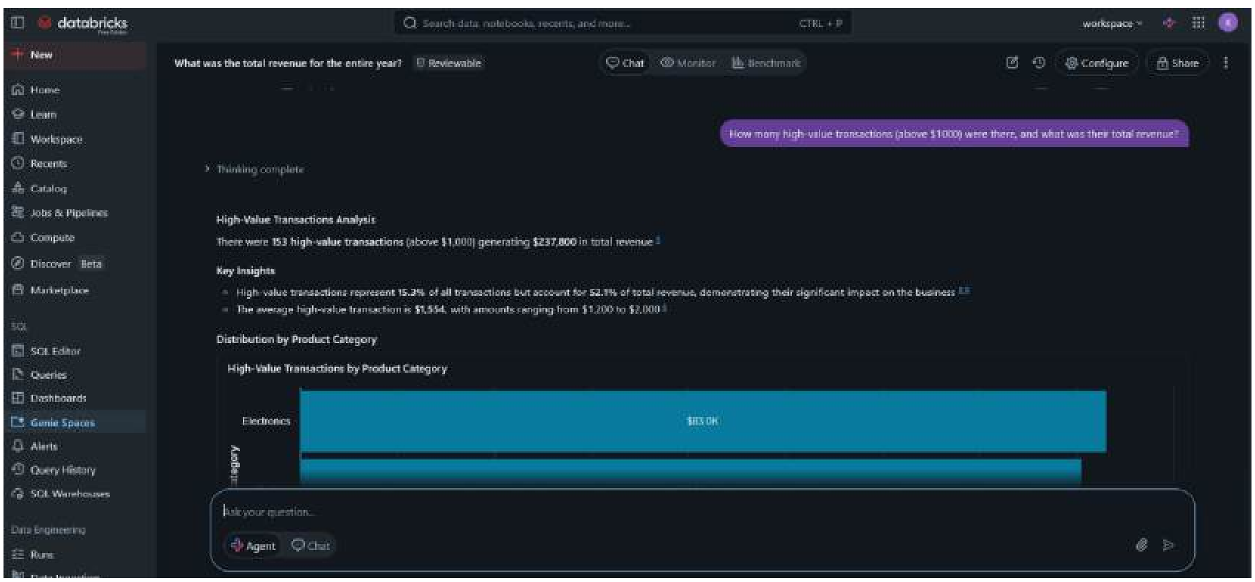
Configure
Share
What is the average age of our customers? Is there a difference in spending between Male and Female customers?
Thinking complete
Customer Age and Gender Spending Analysis
The average customer age is 41.4 years, with customers ranging from 18 to 64 years old
Gender Spending Comparison
There is minimal difference in spending between male and female customers:
Revenue Comparison by Gender
Female
$232.8K
Gender
Male
$223.2K
Ask your question...
Agent
Chat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What was the total revenue for the entire year?
Reviewable
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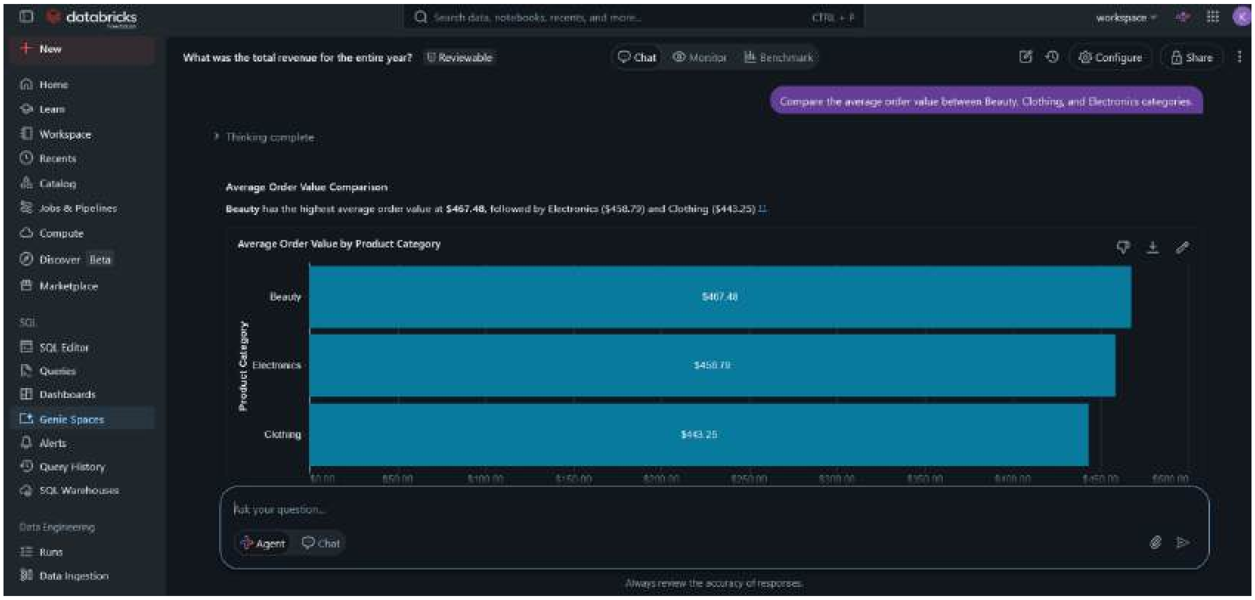
Share
How many high-value transactions (above $1000) were there, and what was their total revenue?
Thinking complete
High-Value Transactions Analysis
There were 153 high-value transactions (above $1,000) generating $237,800 in total revenue
Key Insights
High-value transactions represent 15.3% of all transactions but account for 52.1% of total revenue, demonstrating their significant impact on the business
The average high-value transaction is $1,554, with amounts ranging from $1,200 to $2,000
Distribution by Product Category
High-Value Transactions by Product Category
Electronics
Ask your question...
Agent
Chat


databricks
What was the total revenue for the entire year?
Reviewable
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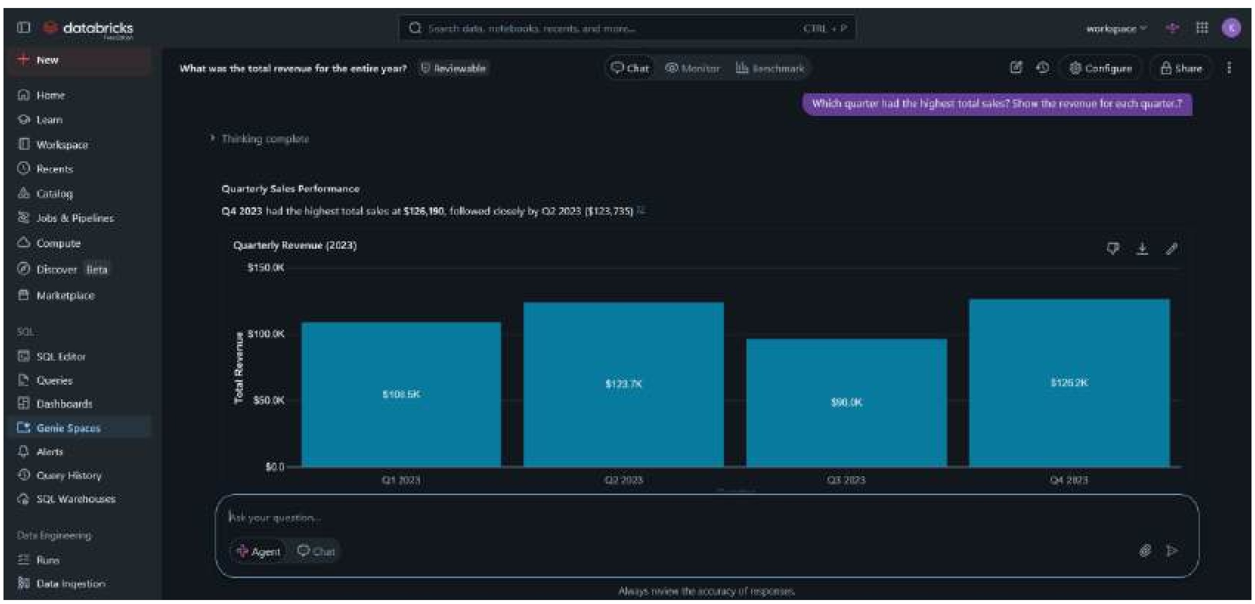
Benchmark
Configure
Share
Compare the average order value between Beauty, Clothing, and Electronics categories.
Thinking complete
Average Order Value Comparison
Beauty has the highest average order value at $467.48, followed by Electronics ($458.79) and Clothing ($443.25)
Average Order Value by Product Category
Product Category
Beauty
$467.48
Electronics
$458.79
Clothing
$443.25
Ask your question...
Agent
Chat
Always review the accuracy of responses.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What was the total revenue for the entire year?
Reviewable
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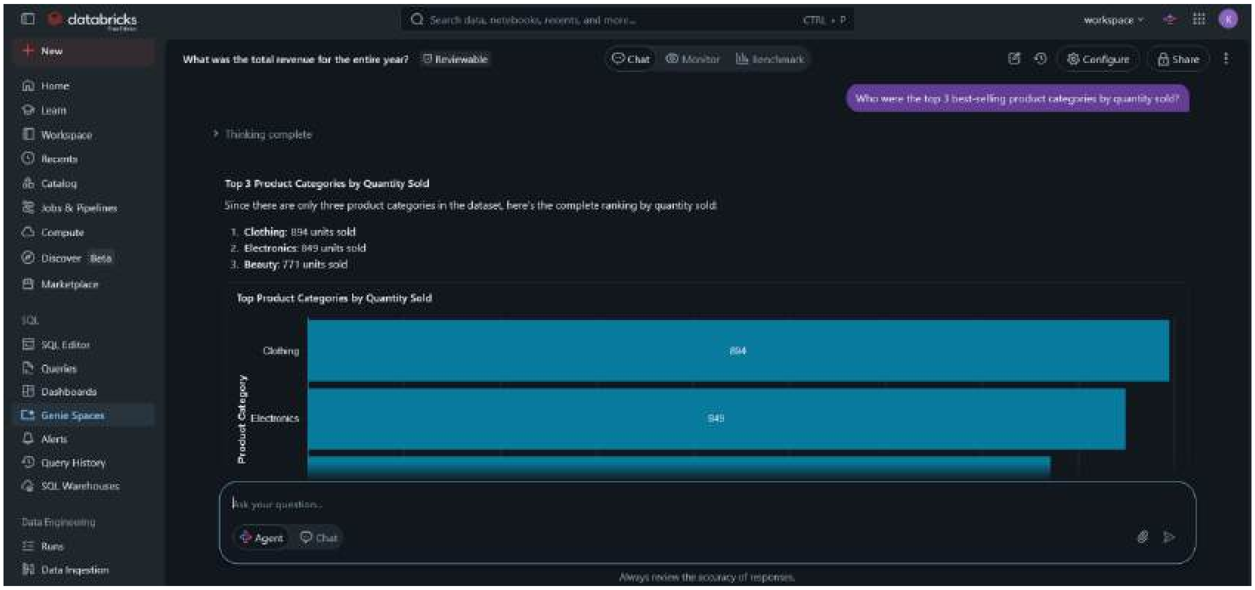
Which quarter had the highest total sales? Show the revenue for each quarter.?
Thinking complete
Quarterly Sales Performance
Q4 2023 had the highest total sales at $126,190, followed closely by Q2 2023 ($123,735)
Quarterly Revenue (2023)
Total Revenue
$150.0K
$100.0K
$50.0K
$0.0
Q1 2023
Q2 2023
Q3 2023
Q4 2023
$123.7K
$126.2K
Ask your question...
Agent
Chat
Always review the accuracy of responses.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What was the total revenue for the entire year?
Reviewable
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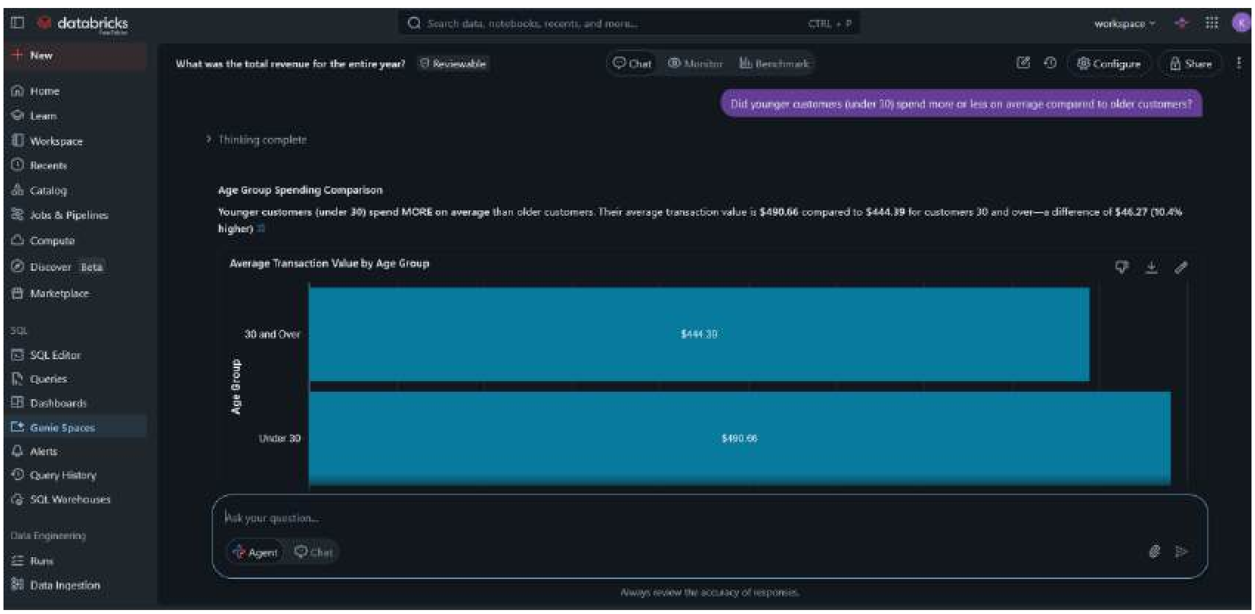
Who were the top 3 best-selling product categories by quantity sold?
Thinking complete
Top 3 Product Categories by Quantity Sold
Since there are only three product categories in the dataset, here's the complete ranking by quantity sold:
1. Clothing: 894 units sold
2. Electronics: 849 units sold
3. Beauty: 771 units sold
Top Product Categories by Quantity Sold
Product Category
Clothing
894
Electronics
849
Ask your question...
Agent
Chat
Always review the accuracy of responses.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What was the total revenue for the entire year?
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Chat
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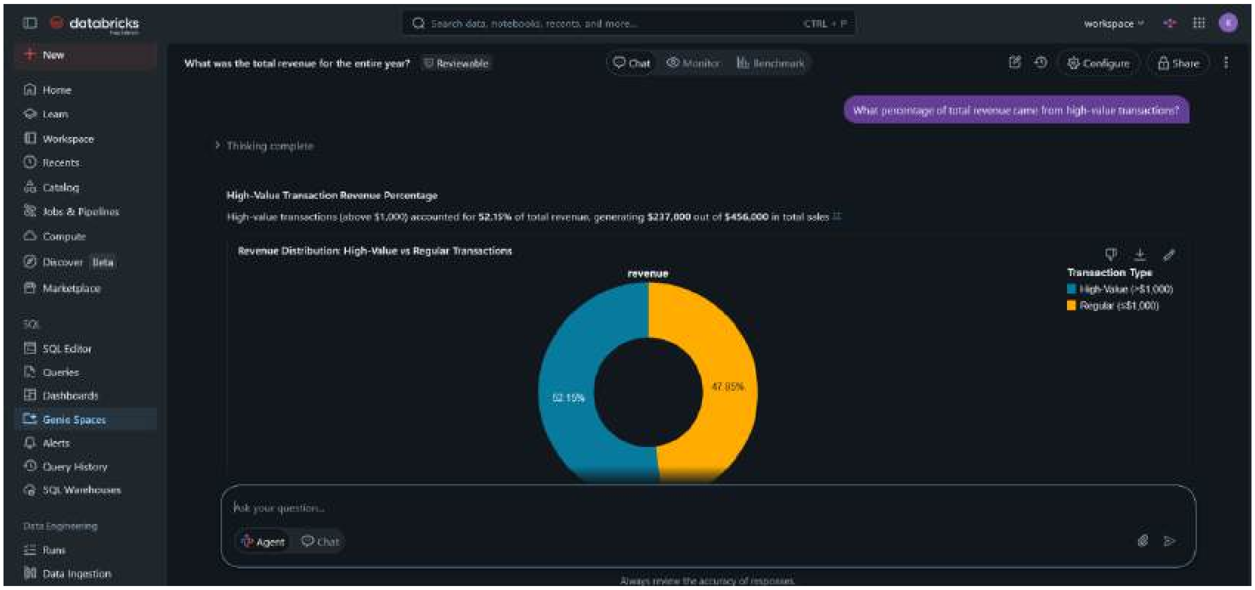
Benchmark
Configure
Share
Did younger customers (under 30) spend more or less on average compared to older customers?
Thinking complete
Age Group Spending Comparison
Younger customers (under 30) spend MORE on average than older customers. Their average transaction value is $490.66 compared to $444.39 for customers 30 and over—a difference of $46.27 (10.4% higher)
Average Transaction Value by Age Group
Age Group
30 and Over
$444.39
Under 30
$490.66
Ask your question...
Agent
Chat
Always review the accuracy of responses.



7. <u>VALIDATE 3</u>

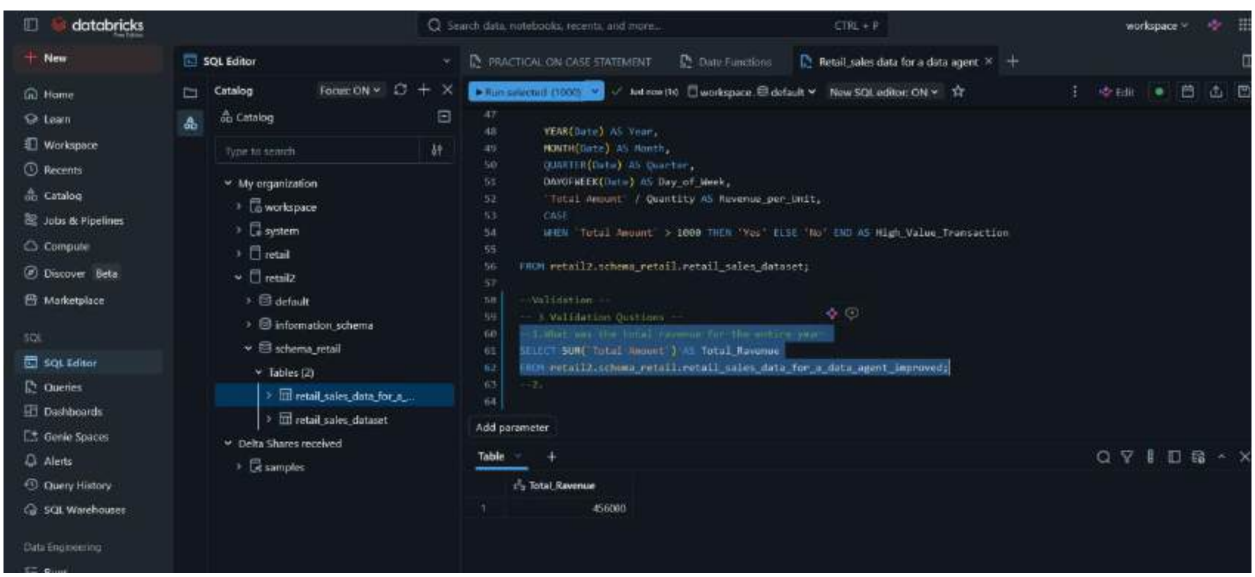
In this step, I chose three questions to validated the Retail Sales Data Agent by testing is with 3 main important questions. The agent's responses were cross-checked using SQL queries.

Validation Results:

1. Question: What was the total revenue for the entire year?

Agent Answer: **$456,000**

Below is my sql query i used to validate the agent's answer

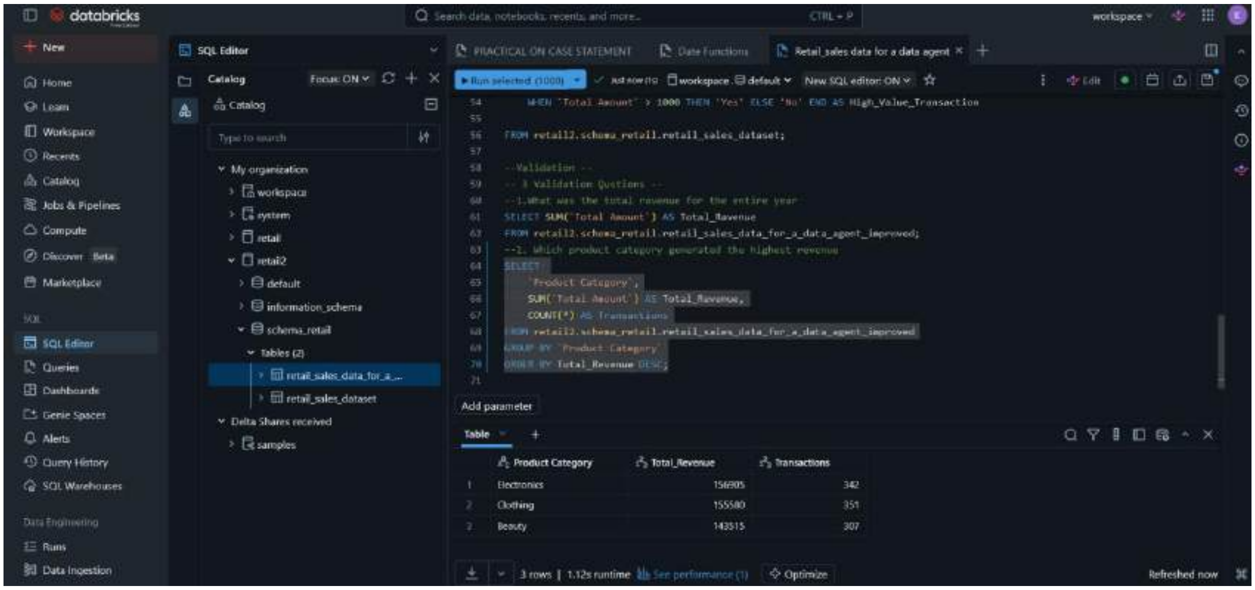


Validation: ✅ Its very much Accurate

2. Question: Which product category generated the highest revenue?

Agent Answer: **Electronics** generated the highest revenue at **$156,905**

My sql query



Validation: ✅ Accurate

3. Question: What is the average age of customers and spending difference between Male and Female?

Agent Answer: Customer Age and Gender Spending Analysis

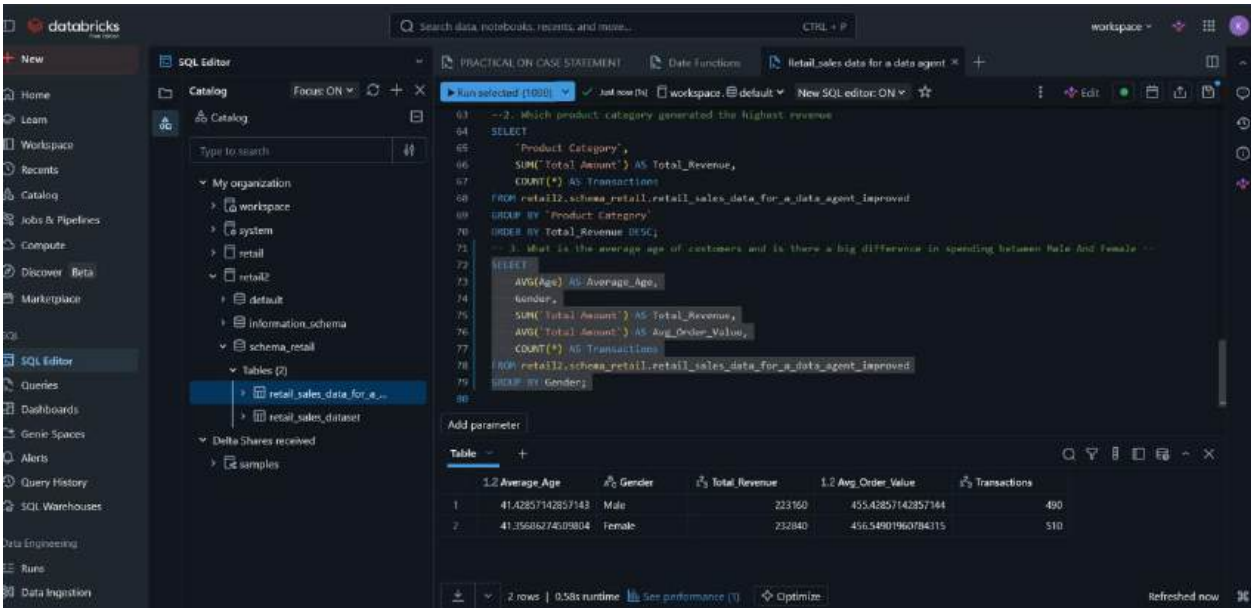
The average customer age is **41.4 years**, with customers ranging from 18 to 64 years old 5

Gender Spending Comparison

There is **minimal difference** in spending between male and female customers:

My sql query:

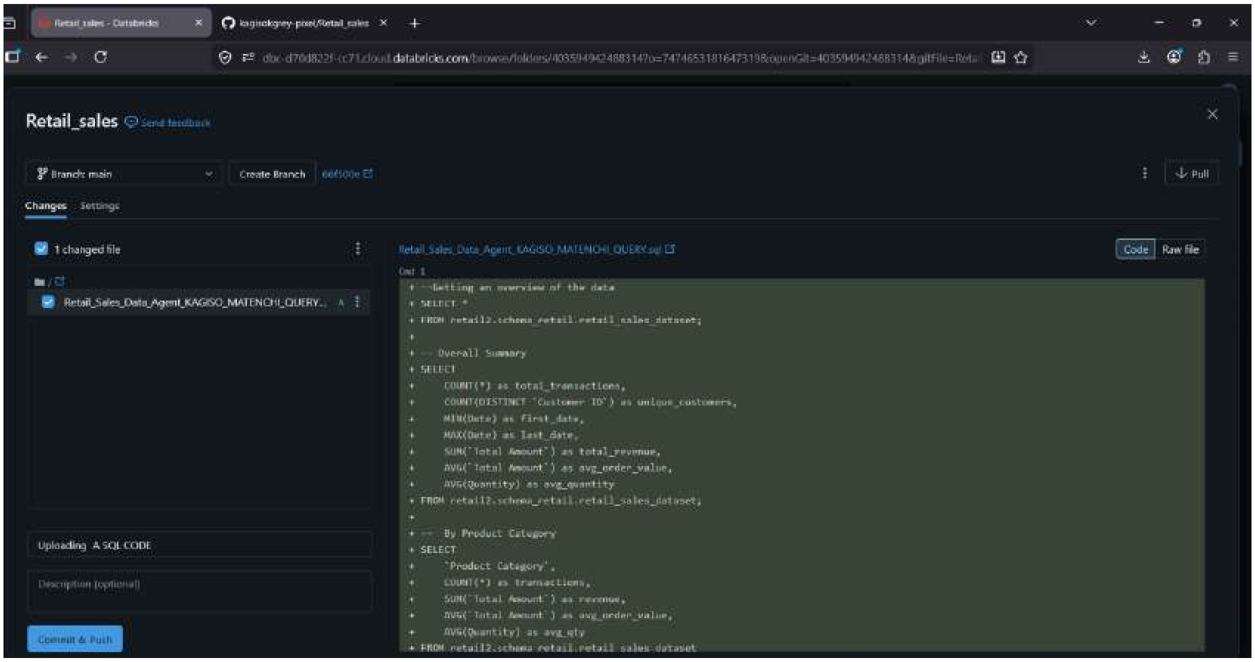
.



Validation: ✅ Accurate

----------------------------THE END--------------------------------

SUBMITING ON GITHUB



Retail_sales - Databricks
kagisokgrey-pixel/Retail_sales
Retail_sales
Send feedback
Branch: main
Create Branch
Pull
Changes
Settings
1 changed file
Retail_Sales_Data_Agent_KAGISO_MATENCHI_QUERY...
Uploading A SQL CODE
Description (optional)
Commit & Push
Retail_Sales_Data_Agent_KAGISO_MATENCHI_QUERY.sql
Code
Raw file
+ --Getting an overview of the data
+ SELECT *
+ FROM retail2.schema_retail.retail_sales_dataset;
+ -- Overall Summary
+ SELECT
+ COUNT(*) as total_transactions,
+ COUNT(DISTINCT `Customer ID`) as unique_customers,
+ MIN(Date) as first_date,
+ MAX(Date) as last_date,
+ SUM(`Total Amount`) as total_revenue,
+ AVG(`Total Amount`) as avg_order_value,
+ AVG(Quantity) as avg_quantity
+ FROM retail2.schema_retail.retail_sales_dataset;
+ -- By Product Category
+ SELECT
+ `Product Category`,
+ COUNT(*) as transactions,
+ SUM(`Total Amount`) as revenue,
+ AVG(`Total Amount`) as avg_order_value,
+ AVG(Quantity) as avg_qty
+ FROM retail2.schema_retail.retail_sales_dataset